Comprehensive System Design Document

Waypoint: AI-Powered Travel Planning & Itinerary Assistant

1. Executive Summary & Problem Statement

Travel discovery platforms accumulate thousands of historical user ratings and reviews. However, existing tools either present flat search results without synthesis or rely on naive LLM wrappers that fabricate itineraries, hallucinate fake hotel ratings, and fail at budget arithmetic. Waypoint solves this by pairing Google Gemini with a rigorous 8-agent LangGraph workflow, hybrid keyword/semantic search over UCI Dataset 485, live OpenStreetMap validation, deterministic budget modeling, and LLM-as-a-judge quality validation with automatic self-refinement.

2. Requirements Matrix

Category	Requirement	Implementation Mechanism
Functional	Natural Language Input	Input Guardrails + Regex entity extraction + Gemini classification.
Functional	Hybrid RAG Retrieval	ChromaDB (dense) + BM25 Okapi (lexical) + Reciprocal Rank Fusion.
Functional	8 Specialized Agents	LangGraph Typed StateGraph with decoupled nodes including Flight Finder.
Functional	Budget Feasibility	Deterministic Python expense allocations & confidence ratio modeling.
Functional	Live Place Grounding	OpenStreetMap (Nominatim & Photon) live attraction and hotel lookup.
Quality	LLM-as-a-Judge	Quality Validator evaluating alignment, feasibility, and grounding.
Quality	Self-Refinement Loop	Conditional LangGraph edge triggering replanning if quality < 0.80.
Frontend	Dual UI Interface	Streamlit client (frontend.py) + React/Vite/TS editorial workspace.

3. Eight-Agent Architecture & Responsibilities

#	Agent Node	Input State	Output Responsibility
1	Preference Classifier	Sanitized Query	Duration, budget, interests, traveler type, travel style.
2	Destination Analyzer	Preferences, Hybrid RAG	Reranked candidate destinations, match score, evidence.
3	Budget Analyst	Destination, Duration	Itemized expenses (Stay 38%, Food 24%, Activities 18%, Transit 12%, Misc 8%).
4	Activity Recommender	Interests, Profile	Top activities with time slots, cost estimates, and ratings.
5	Accommodation Finder	Stay Budget, Style	Grounded lodging tiers, features, strategic location.
6	Flight Finder	Origin, Destination, Budget	Flight options across budget/standard/premium with realistic fares.
7	Itinerary Planner	All Prior Agents	Day-by-day Morning/Afternoon/Evening chronological schedule.
8	Quality Validator	Final Itinerary & State	LLM Judge quality score; triggers refinement loop if < 0.80.

4. Recommendation Scoring Formula

Candidate destinations are ranked using a multi-factor transparent formula:
Final Score = w1·PreferenceMatch + w2·SemanticRelevance + w3·HistoricalRating + w4·EvidenceStrength + w5·BudgetCompatibility
Default Weights: w1=0.30, w2=0.25, w3=0.20, w4=0.15, w5=0.10. All components are normalized between 0.0 and 1.0.

5. Trade-offs & Design Decisions

• **Deterministic Calculations vs. LLM Arithmetic:** Budget allocations and currency math are strictly computed via Python routines to avoid LLM hallucination in numeric arithmetic.
• **OpenStreetMap Integration vs. Static Memorization:** Combines local high-speed cache with live OpenStreetMap Photon API so even newly discovered cities have real, verifiable tourist sights.
• **Local Persistent Embeddings:** ChromaDB runs locally without external cloud dependencies, guaranteeing low latency, zero external API costs for retrieval, and 100% offline test compliance.
• **Dual Frontend Interface:** Ships with both a fast Streamlit client (runnable via *streamlit run frontend.py*) and an enterprise React/Vite dashboard to meet both evaluator submission specs and modern web standards.